SOP: Configure Point-to-Site (P2S) VPN in Microsoft Azure

Objective

To configure a Point-to-Site (P2S) VPN connection in Microsoft Azure using Azure Certificate Authentication, enabling secure connectivity between a local client machine and an Azure Virtual Network (VNet). The VPN connection was validated by accessing a private Azure Linux Virtual Machine through its private IP address.

Prerequisites

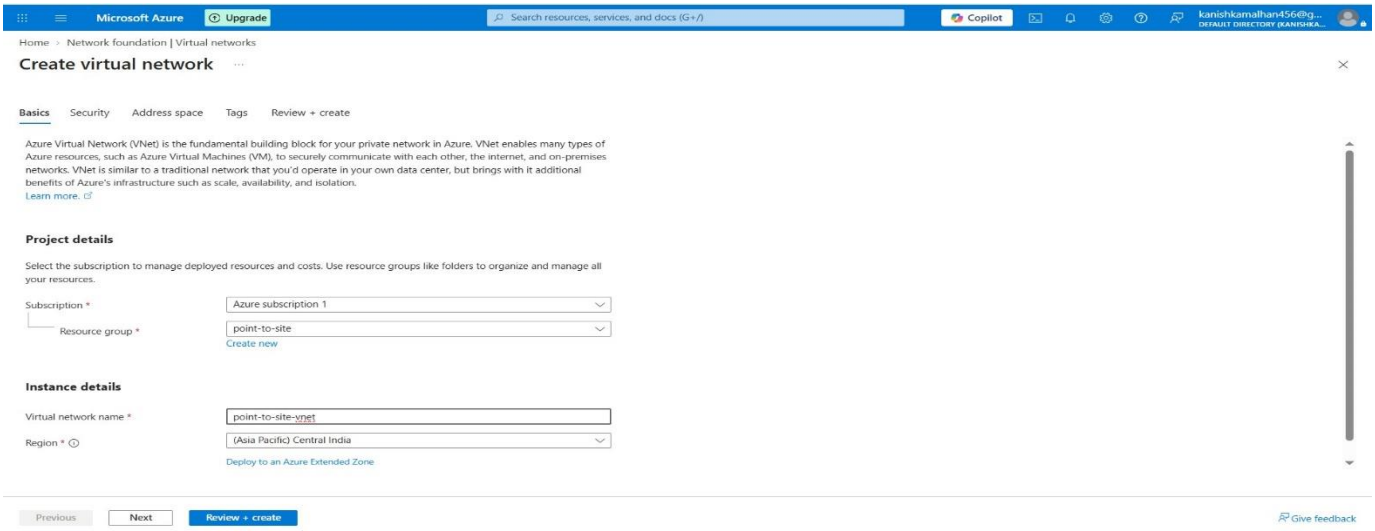
- Azure Subscription
- Azure Resource Group
- Azure Virtual Network (VNet)
- Azure Virtual Network Gateway
- Azure VPN Client
- PowerShell
- Windows Client Machine

Step 1: Create Resource Group

Created a dedicated Resource Group to host all Point-to-Site VPN resources.

Configuration

Resource Group Name: point-to-site
Region: Central India



Step 2: Create Virtual Network

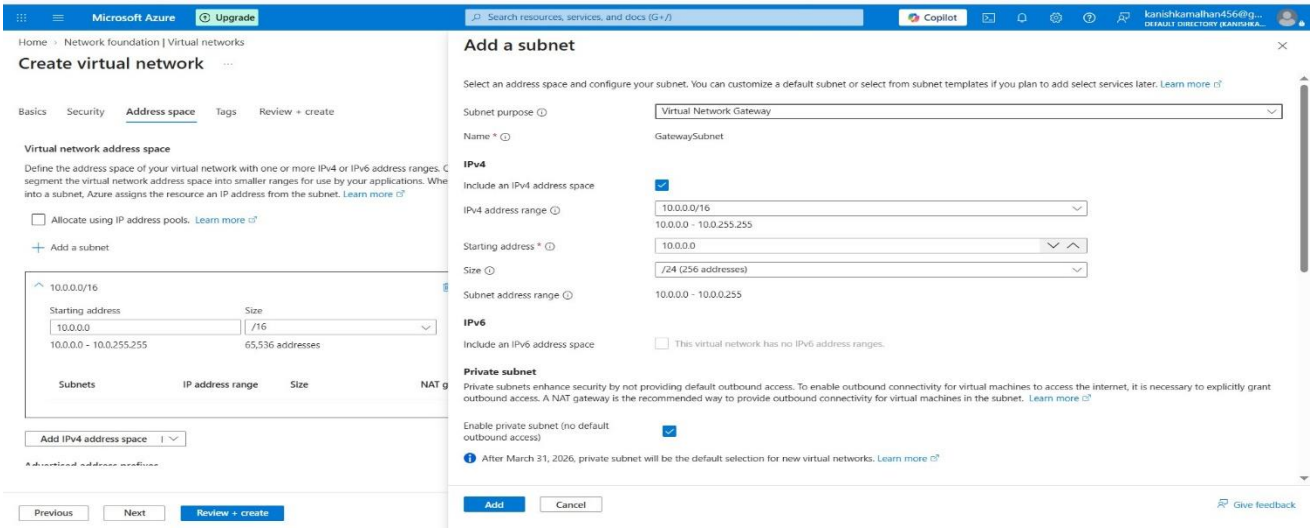
Created a Virtual Network to host the Virtual Machine and VPN infrastructure.

Configuration

VNet Name: point-to-site-vnet
Address Space: 10.0.0.0/16
Region: Central India

Purpose

The Virtual Network serves as the private network in which Azure resources communicate securely.



Step 3: Create Required Subnets

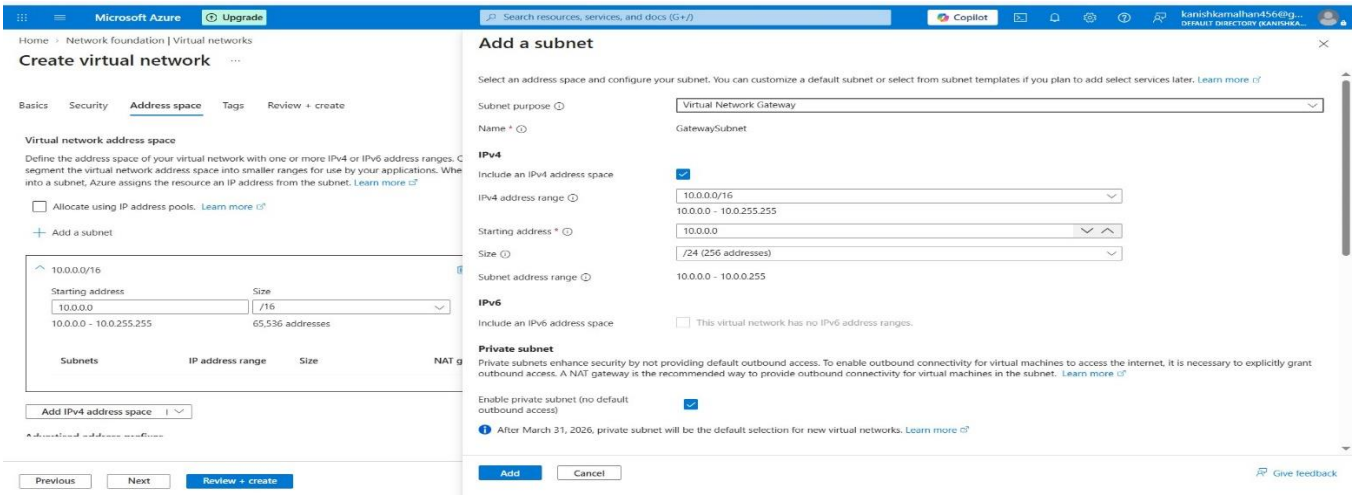
Configured two subnets inside the Virtual Network.

VM Subnet

Used for hosting the Linux Virtual Machine.

GatewaySubnet

Used exclusively for the deployment of the Virtual Network Gateway.

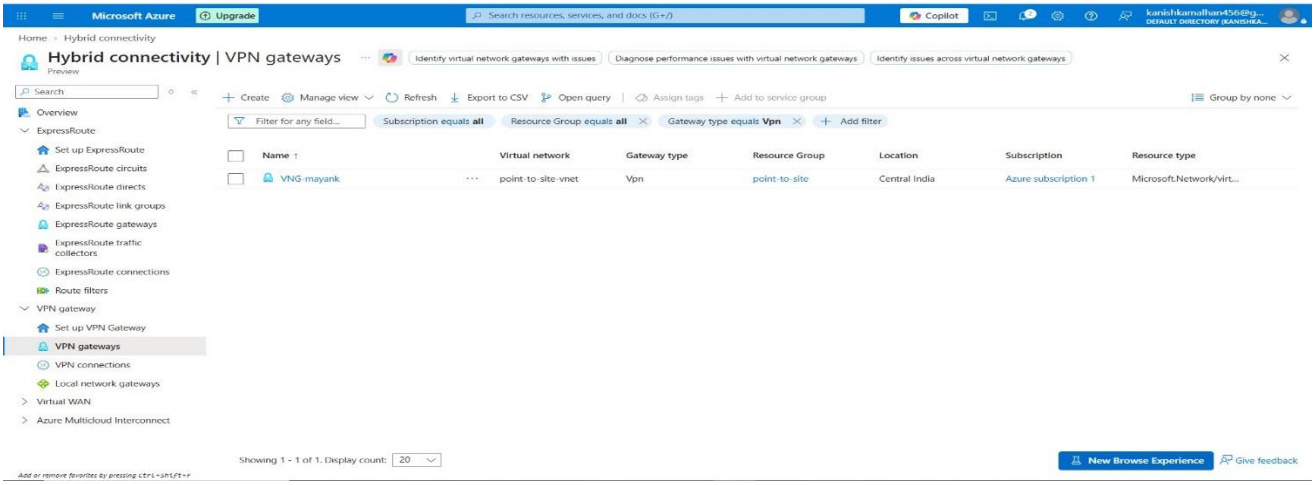


Step 4: Deploy Virtual Network Gateway

Created a Virtual Network Gateway to provide secure VPN connectivity.

Purpose

The VPN Gateway acts as the Azure endpoint responsible for establishing encrypted VPN tunnels between the on-premises client and Azure Virtual Network.



Step 5: Generate Root Certificate

Generated a self-signed Root Certificate using PowerShell.

Purpose

The Root Certificate acts as the trusted certificate authority for VPN client authentication.

Outcome

A Root Certificate named:
P2SRootCert

Administrator: Windows PowerShell

```
> }
> $cert = New-SelfSignedCertificate @params
S C:\Windows\system32> $params = @{
>         Type = 'Custom'
>         Subject = 'CN=P2SChildCert'
>         DnsName = 'P2SChildCert'
>         KeySpec = 'Signature'
>         KeyExportPolicy = 'Exportable'
>         KeyLength = 2048
>         HashAlgorithm = 'sha256'
>         NotAfter = (Get-Date).AddMonths(18)
>         CertStoreLocation = 'Cert:\CurrentUser\My'
>         Signer = $cert
>         TextExtension = @(
>           '2.5.29.37={text}1.3.6.1.5.5.7.3.2')
>     }
>     New-SelfSignedCertificate @params


  PSParentPath: Microsoft.PowerShell.Security\Certificate::CurrentUser\My

humbprint                                Subject
---------                                -------
35774235B40072D3362BD674AC40A0F83431EBA  CN=P2SChildCert


S C:\Windows\system32>
```

Step 6: Generate Child (Client) Certificate

Generated a Child Certificate signed by the Root Certificate.

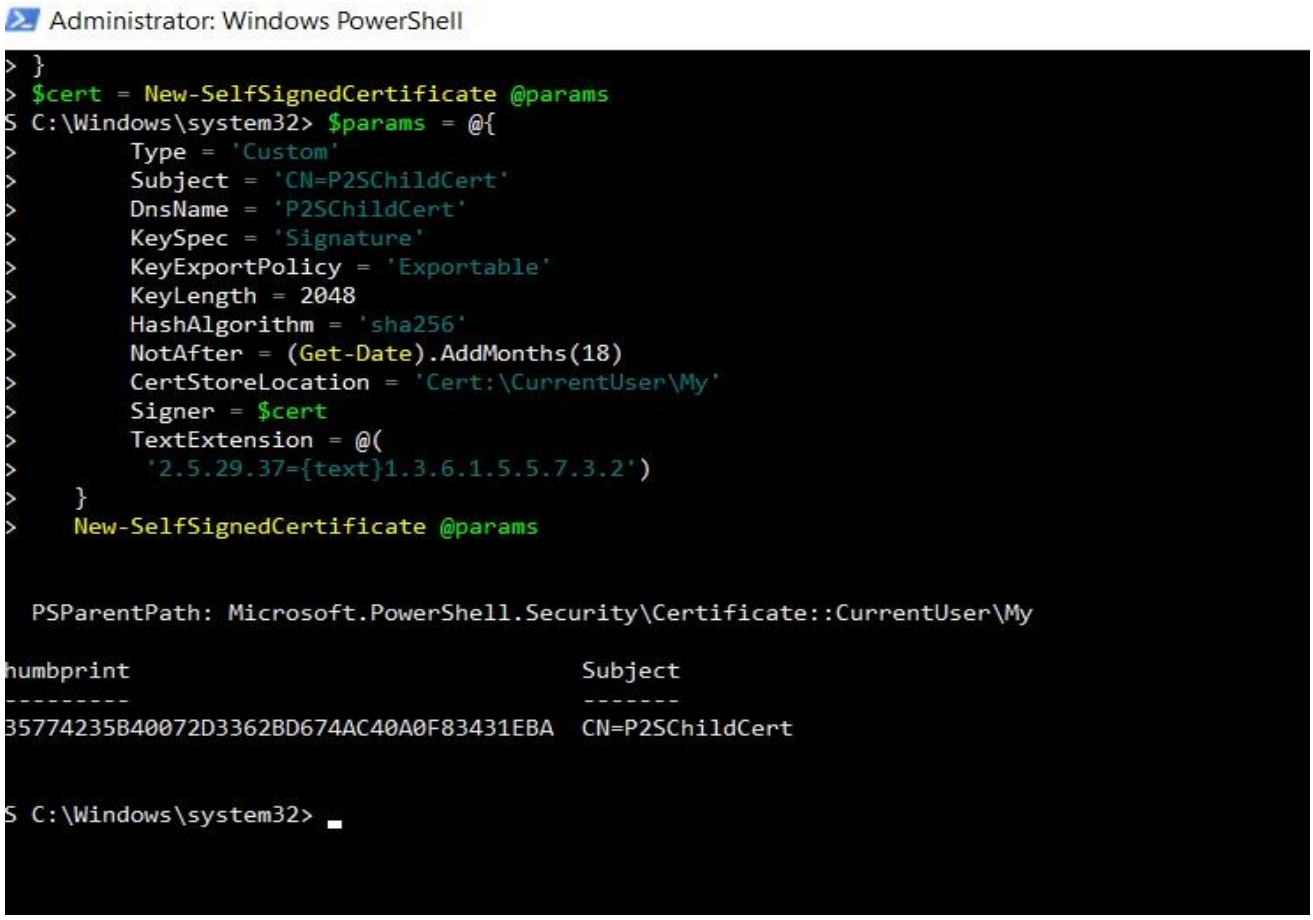
Purpose

The Client Certificate is used by Azure VPN Client during the authentication process.

Certificate Name: P2SChildCert

Outcome

A Client Authentication certificate was successfully generated and signed by the Root Certificate.

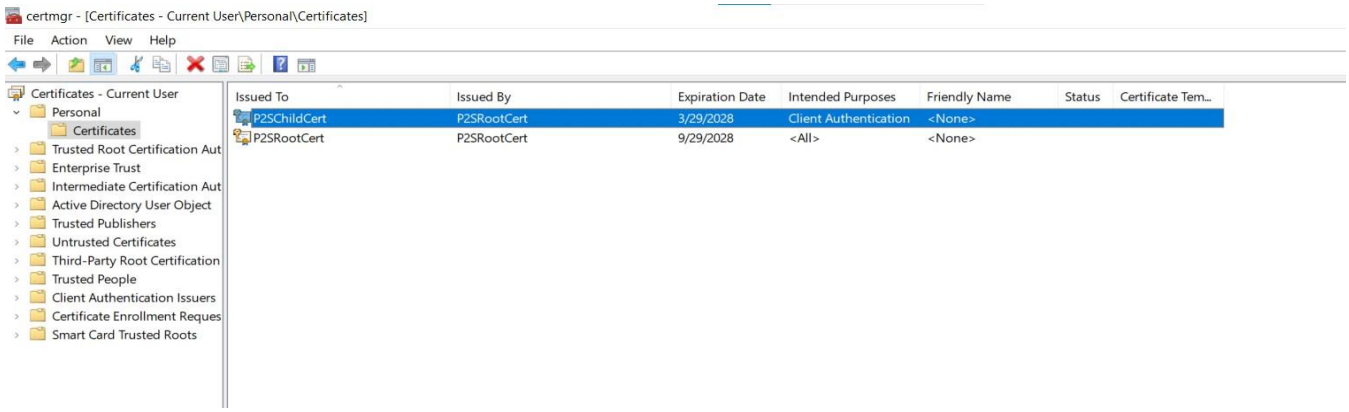


Step 7: Verify Certificates in Local Certificate Store

Validated that both certificates were successfully available in the client machine certificate store.

Certificates Verified

P2SRootCert
P2SChildCert



Step 8: Export Certificates

Exported both Root and Child Certificates.

Exported Files

rootcertificate.cer
clientcertificate.pfx

Purpose

- Root Certificate uploaded to Azure VPN Gateway.
- Child Certificate installed on the client machine.



Step 9: Configure Point-to-Site VPN

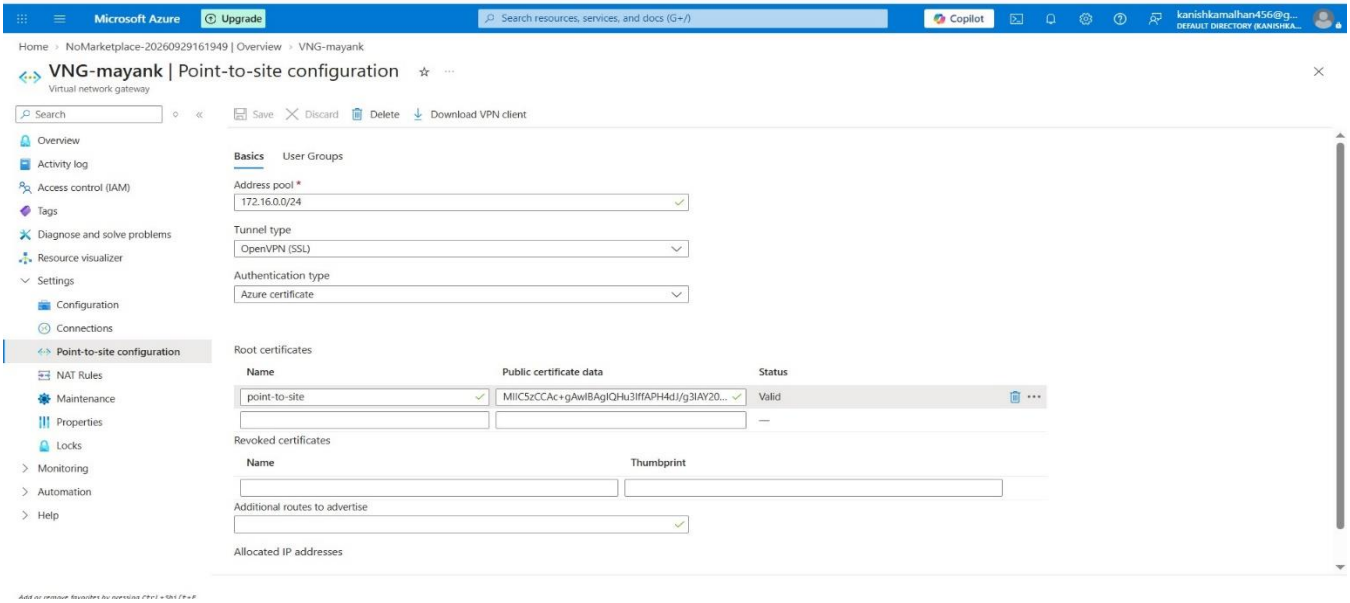
Configured Point-to-Site settings on the Virtual Network Gateway.

Configuration

Address Pool: 172.16.0.0/24
Tunnel Type: OpenVPN (SSL)
Authentication Type: Azure Certificate

Activities Performed

- Added Root Certificate
- Uploaded Public Certificate Data
- Validated Certificate Status
- Saved Point-to-Site Configuration



Step 10: Download VPN Client Package

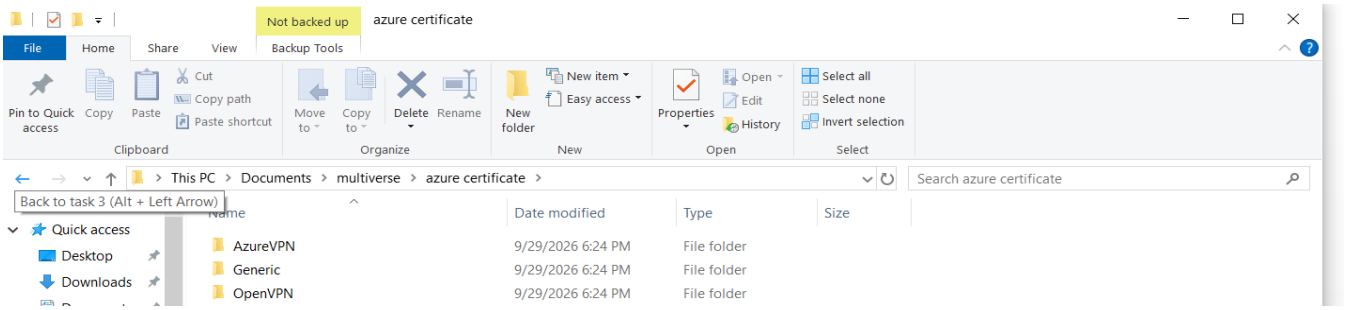
Downloaded the VPN Client package from Azure Portal after completing Point-to-Site VPN configuration.

Purpose

The downloaded VPN package contains the Azure VPN Client profile required for establishing VPN connectivity.

Activities Performed

- Downloaded VPN Client Package
- Extracted VPN Configuration Files



Step 11: Install and Configure Azure VPN Client

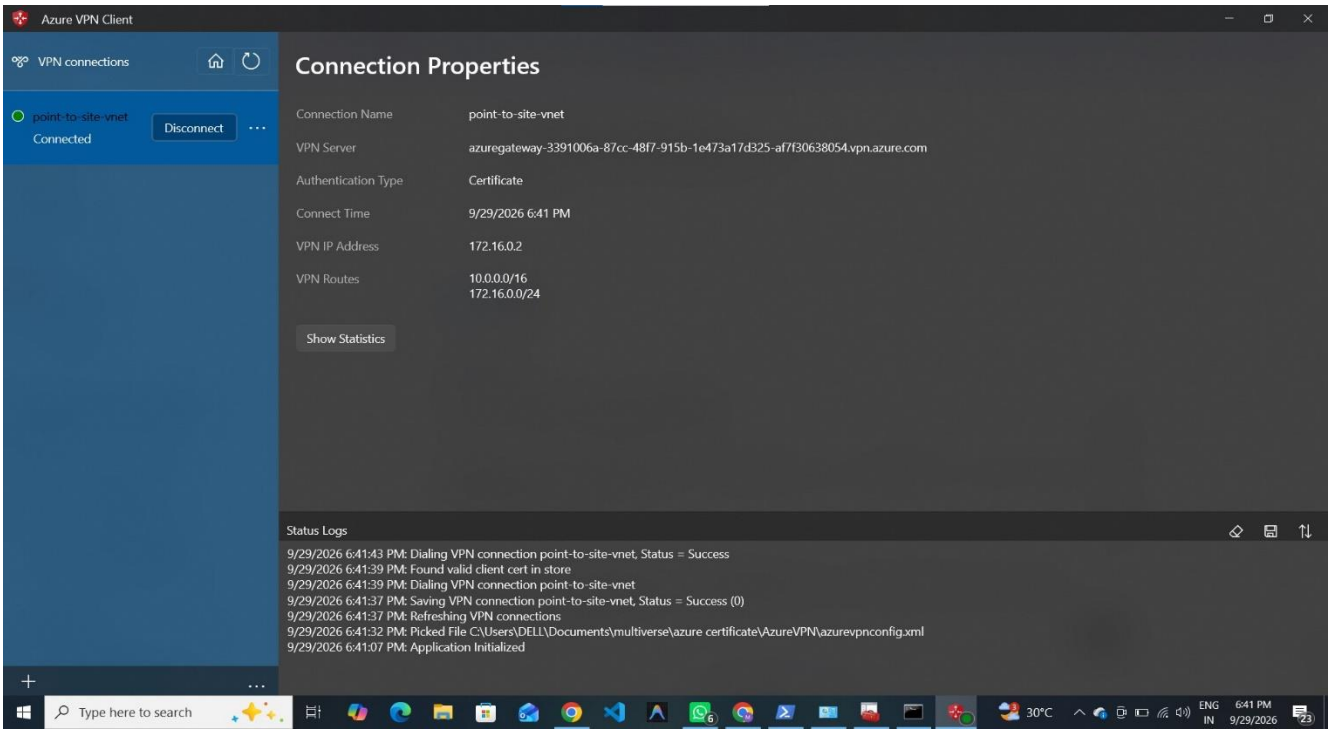
Installed Azure VPN Client and imported the downloaded VPN profile configuration.

Activities Performed

- Installed Azure VPN Client
- Imported Azure VPN Configuration File
- Selected Child Certificate (P2SChildCert)
- Saved VPN Profile

Purpose

Azure VPN Client is required to establish secure connectivity between the local machine and Azure Virtual Network.



Step 12: Establish VPN Connection

Successfully established the Point-to-Site VPN connection.

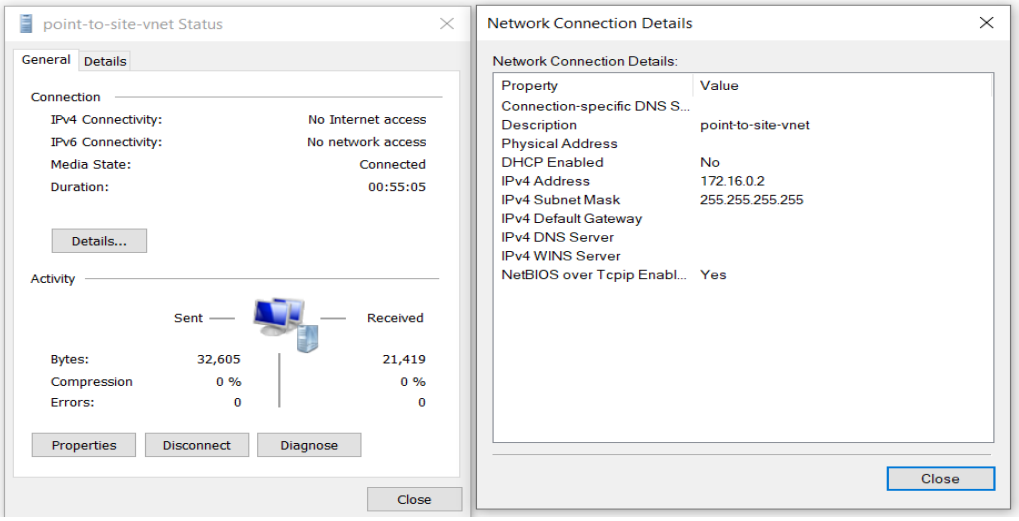
Connection Details

VPN Profile: point-to-site-vnet
Authentication Type: Certificate
Assigned VPN IP: 172.16.0.2

Result

VPN tunnel was established successfully and the client machine received an IP address from the configured VPN address pool.



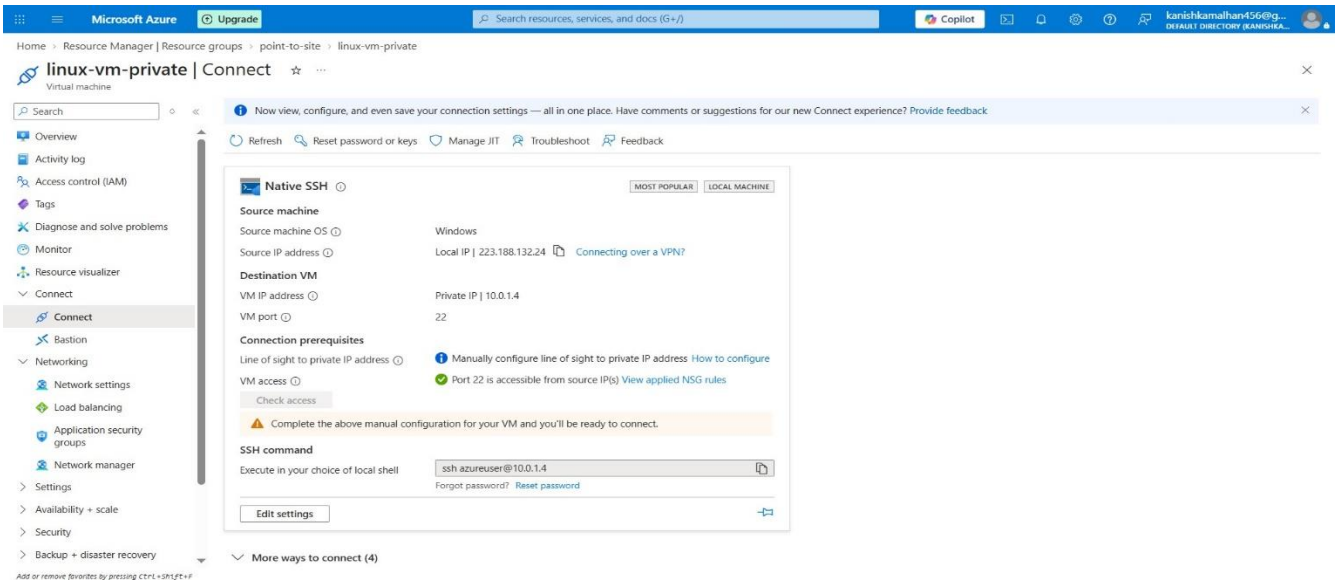


Step 13: Deploy and Validate Private Linux Virtual Machine

A Linux Virtual Machine was deployed within the Virtual Network.

VM Details

VM Name: linux-vm-private
Private IP Address: 10.0.1.4

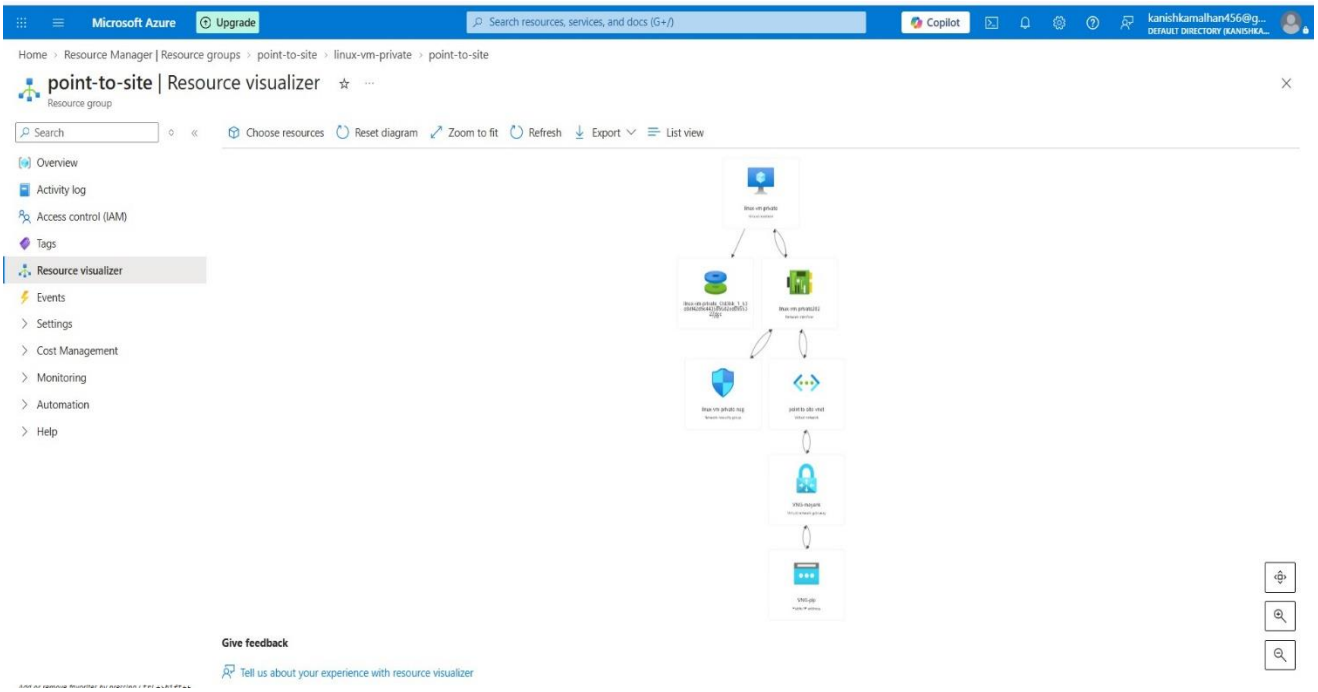


Step 14: Verify Network Topology

Validated the deployed infrastructure using Azure Resource Visualizer.

Verified Components

- Resource Group
- Virtual Network
- Virtual Machine
- Network Interface
- Public IP
- VPN Gateway



Step 15: Validate Access Through Private Network

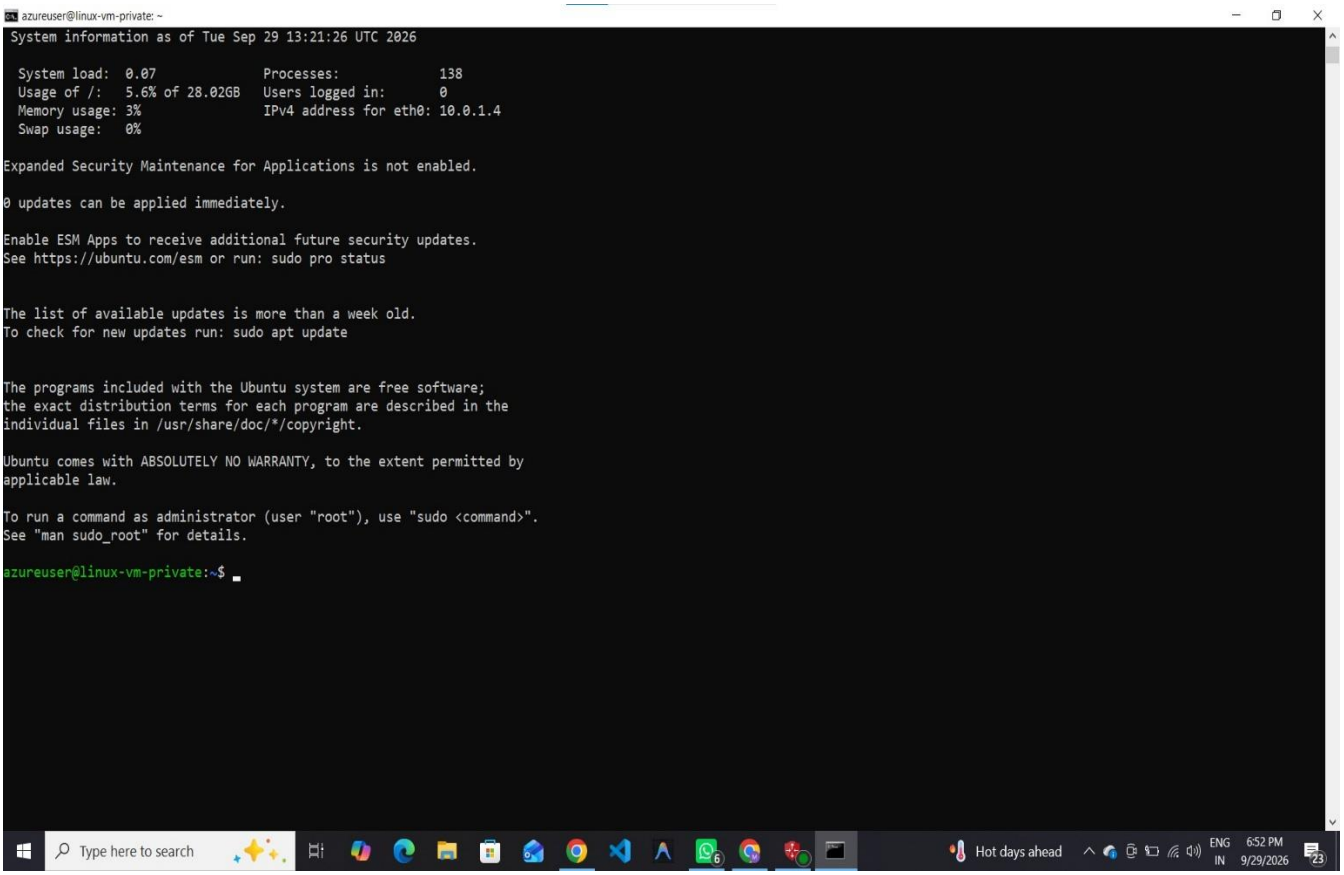
After the VPN connection was established, the Linux Virtual Machine was accessed successfully using its private IP address.

SSH Command

ssh azureuser@10.0.1.4

Validation Performed

- Successfully connected to the Linux VM using Private IP.
- Verified VM accessibility over the VPN tunnel.



Step 16: Validate Web Access Through Private IP

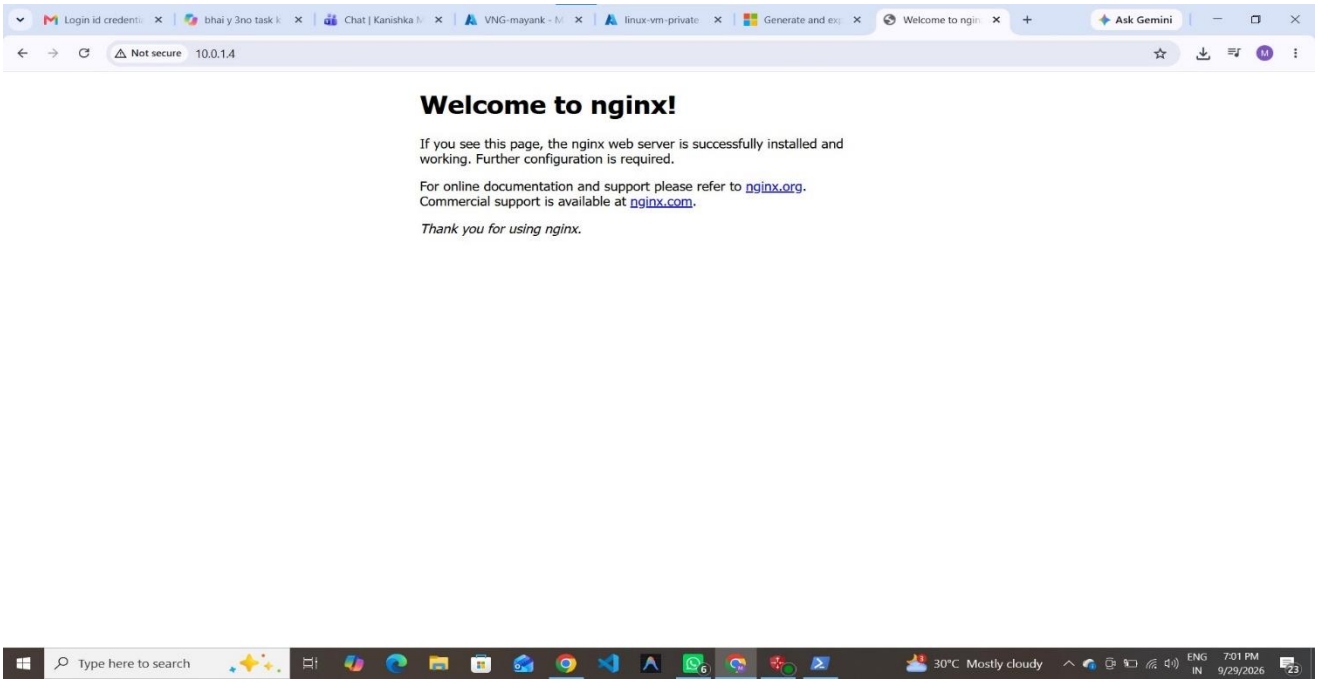
To further confirm connectivity, the NGINX web server hosted on the Linux Virtual Machine was accessed through its private IP address.

URL Accessed

http://10.0.1.4

Result

The default NGINX welcome page was displayed successfully, confirming that the client machine could access services hosted on the Azure VM through the Point-to-Site VPN connection.



Observations

- Resource Group was created successfully.
- Virtual Network and subnets were configured successfully.
- VPN Gateway deployment completed successfully.
- Root and Child Certificates were generated successfully.
- Certificates were exported and validated successfully.
- Point-to-Site VPN configuration was completed successfully.
- Azure VPN Client was configured successfully.
- VPN connection was established successfully.
- VPN client received IP address **172.16.0.2** from the configured VPN address pool.
- Linux Virtual Machine was reachable through its private IP **10.0.1.4**.
- SSH connectivity over VPN was successful.
- NGINX web application was accessible through the private IP over the VPN connection.

Result

Successfully configured and validated an Azure Point-to-Site (P2S) VPN using Azure Certificate Authentication. A secure VPN tunnel was established between the local workstation and Azure Virtual Network through Azure VPN Client. The client machine received VPN IP address **172.16.0.2** and successfully accessed the Azure Linux Virtual Machine **(10.0.1.4)** using both SSH and a web browser over the private network, confirming successful Point-to-Site VPN implementation.